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Chladni-Inspired Aesthetic Patterns: A Flexible Parametric Model from Thin-Plate Vibrations

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***Abstract*—We introduce a physics-based parametric generative model for a class of wave-like, organic aesthetic patterns that streamlines the conventional two-stage design workflow. Unlike conventional approaches that rely on separate texture design and manual invariant mappings for seamless tiling, our method simultaneously derives texture and layout from the explicit solution of the thin-plate vibration equation. By encoding symmetry and tiling constraints directly as boundary conditions, the model inherently guarantees seamless, symmetric patterns without ad-hoc mapping functions. Within this framework, artists can intuitively control style and structure through a compact set of visually interpretable parameters, enabling efficient generation of diverse periodic and aperiodic tilings. Extensive experiments validate the method's efficiency, structural coherence, and stylistic richness, demonstrating clear advantages over conventional approaches in both visual diversity and structural integrity. This work offers a novel, interpretable, and efficient approach to controllable digital art creation.**

***Index Terms*—Aesthetic Patterns, Thin Plate Vibrations, Mathematical Visualization, Symmetrical Layout, Tiling Structures.**

I. Introduction

AESTHETIC patterns are ubiquitous in modern design [1], [2], with applications spanning textiles, interior decoration, and architectural surfaces. A rich ecosystem of computational techniques has emerged for their generation, including fractal geometry, dynamical systems, Wang tiles, procedural arrangements, and Artificial Intelligence (AI)-driven models [3]. Among these, physics-based approaches are particularly appealing: the visual richness of Chladni patterns, standing wave nodal lines revealed by sand on vibrating plates, has long captivated artists and scientists alike [4], inspiring a tradition of cymatic art that motivates our work.

Existing approaches to pattern generation broadly fall into two categories: those that generate texture content via nonlinear theories (e.g., fractals [5], dynamical systems (DS) [6], and quasi-regular patterns (QRP) [7]), and those that structure spatial layout using symmetric functions [8], tiling templates [9], or complex-plane methods [10], [11]. A persistent challenge, however, is the separation of texture generation from layout control. Tiling-based methods, in particular, depend on manually constructed invariant mappings (IMs) [1], [12]–[14] to ensure seamless transitions; this labor-intensive and structure-dependent process limits model flexibility and generalizability.

To overcome this limitation, we introduce a physics-inspired parametric model based on the Thin-Plate Free Vibration Equations (TPFVEs). Rather than aiming for universal coverage, we focus on a specific class of wave-like, organic textures characterized by undulating contours and symmetric tiling structures. Our core innovation is a unified Partial Differential Equation (PDE) formulation in which both pattern content and structural constraints are encoded simultaneously, with symmetry and seamless tiling imposed directly as boundary conditions. This eliminates the need for ad-hoc IMs and enables intuitive control through a compact set of visually interpretable parameters (e.g., texture density, aspect ratio, and motion). The method naturally accommodates both periodic and aperiodic tilings, ensuring structural integrity and visual continuity by construction. While our approach is designed specifically for wave-like aesthetic patterns with symmetric tiling structures, rather than as a replacement for existing paradigms, it offers a flexible, interpretable, and computationally efficient alternative within this well-defined domain. Extensive experiments confirm its ability to generate a diverse range of rich, high-quality patterns, providing a principled and controllable pathway for digital art creation.

The remainder of this paper is organized as follows. Sec. II reviews related work. Sec. III introduces the thin-plate vibration model and derives its general solution. Sec. IV presents boundary condition formulations for symmetry and tiling continuity. Sec.V describes the GPU-accelerated rendering pipeline. Sec. VI reports experimental results and comparisons. Section VII concludes the paper and discusses future directions.

II. Related work

Aesthetic pattern generation spans diverse techniques. Gieseke et al. [3] provide a comprehensive taxonomy of control mechanisms. Beyond nonlinear methods discussed below, influential paradigms include Wang tiles for seamless non-periodic textures [15], procedural element arrangement [16], and AI-driven generative models [17]. This work focuses on a physics-driven PDE formulation that inherently couples content generation and spatial layout.

A. Methods based on nonlinear theory

A substantial body of work builds upon nonlinear theories, notably fractal geometry [5], dynamical systems [14], and quasi-regular patterns [1].

Fractal patterns exploit self-similarity to generate complex structures from compact contraction mappings. Iconic examples include the Mandelbrot and Julia sets [18], which are

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rendered via escape-time, orbit-trap, or root-finding methods [8], [19], [20]. However, fractal details strictly follow mathematical rules, limiting local control, and the sensitivity to initial parameters introduces design uncertainty.

Dynamical systems describe state evolution through differential equations or iterative mappings. Szyszkowicz [6] pioneered their use for digital pattern generation. While these systems yield regular, coherent structures with adjustable complexity, their chaotic nature makes systematic control difficult, as small perturbations can lead to unpredictable diverging outcomes. Moreover, under certain conditions, their outputs may appear overly regular or repetitive.

Quasi-regular patterns arise from visualizing the height field of functions derived from the ZZSUC (Zaslavsky–Zakharov–Sagdeev–Usikov–Chernikov) map [21], with transformations and smoothing. Zhang et al. [22] first proposed this visualization method, which was later extended [7], [23]. These patterns exhibit strong regularity, symmetry, and stability, making them suitable for textile design; however, the limited set of variable parameters restricts stylistic diversity.

Physics-based simulation offers another generative avenue. Jin et al. [24] proposed a GPU-based system that simulates dye advection in viscous fluid to produce marbling textures in real time. Lu et al. [25] extended this approach to creative modeling with user-controlled flow fields. While these methods produce visually rich, fluid-like textures, they are tailored specifically to marbling and do not readily generalize to arbitrary tiling structures or controlled symmetry layouts.

B. Methods incorporating symmetry and tiling

Symmetry and tiling have been widely adopted in pattern generation to enhance aesthetic control. Existing approaches can be grouped into several distinct categories.

Symmetric dynamical systems and complex-plane methods. Chung et al. [26] used Fourier series to generate patterns for the 17 wallpaper groups. Subsequent works extended this direction by constructing various symmetric functions and coloring schemes [27]–[30]. However, these approaches often produce relatively simplistic spatial structures and rely heavily on manual intervention, requiring strong mathematical expertise.

Other methods explore different mathematical routes. Farris [10] employed domain coloring in the complex plane to generate seamless wallpaper patterns; Field and Golubitsky [11] constrained dynamical systems via chaotic attractors to exhibit prescribed symmetry groups; and McCabe [31] used reaction-diffusion to produce cyclic symmetric Turing patterns. Jones and Reiter [32] systematically investigated chaotic attractors with cyclic symmetry, while Dumont and Reiter [33] explored attractors near forbidden symmetry, revealing rich pattern transitions. Gontar and Grechko [34], [35] developed a difference equation system derived from chemical reaction discrete chaotic dynamics (CRDCD), generating symmetric patterns and fractal parameter sets through visualization of concentration fields. Despite their individual strengths, these methods share common limitations: they depend on explicit function composition or system design, offer limited control over arbitrary tile boundaries, and are often restricted to specific symmetry groups.

Shape grammar and rule-based layout. Shape grammars provide a declarative approach to generating symmetric geometric patterns [36], [37], enabling high-level design control via recursive production rules. However, they require manual grammar design and do not offer a continuous physical mechanism for texture variation.

Tiling templates and invariant mappings. Another line of work employs tiling templates for spatial composition. Chung et al. [12] generated patterns over aperiodic tilings (Penrose [38], chair, sphinx) by constructing IMs on the fundamental region. Similar mapping-based methods were developed for chair tilings [39], fractal, Archimedean, and spiral tilings [9], [13], [40], as well as k-uniform tilings [1], arbitrary quadrilateral tilings [14], and isohedral tilings [41]. A common limitation is the need for explicit mathematical construction of IMs tailored to specific fundamental regions, which demands manual derivation and restricts generality. Some extensions avoid IMs by adapting Escher-like tilings to regular polyhedra, hyperbolic space, or rosette shapes [2], [42], though these remain most suitable for textured-tile scenarios.

Wang tiles and procedural arrangements. Wang tiles [15] assemble a small set of edge-colored square tiles to generate seamless non-periodic textures. Procedural element arrangement [16] distributes vector graphics primitives according to programmable rules. However, neither method simultaneously controls texture detail and structural continuity within a unified model.

C. Methods driven by AI

Recent advances in deep learning have transformed digital art. Pang et al. [43] proposed a neural network aided design (NAD) framework incorporating design constraints such as color, geometry, and symmetry. Sun et al. [44] combined Markov random fields with the Gram method to accelerate generation while preserving local texture authenticity. Yar et al. [45] introduced TexGAN, a deep convolutional GAN trained on over 17,000 images. Izzuddin et al. [46] developed a prompt-conditioned framework for batik pattern generation using a LoRA-weighted latent diffusion model. Ding et al. [47] presented Text2Printing, a diffusion-based framework with attention modulation that enforces color-keyword binding. Very recently, Aigerman and Groueix [17] proposed Generative Escher Meshes, an AI-driven approach that synthesizes symmetric tilings from user prompts. Despite these advances, AI-based methods remain constrained by dataset scale and quality, heavy computational requirements, and unresolved copyright issues, limiting their applicability for real-time, resource-efficient, and interpretable pattern design.

III. Overview of the Proposed Method

Existing methods for generating aesthetic patterns (e.g., fractals, dynamical systems, quasi-regular patterns) share a core limitation: they require predefining an explicit function

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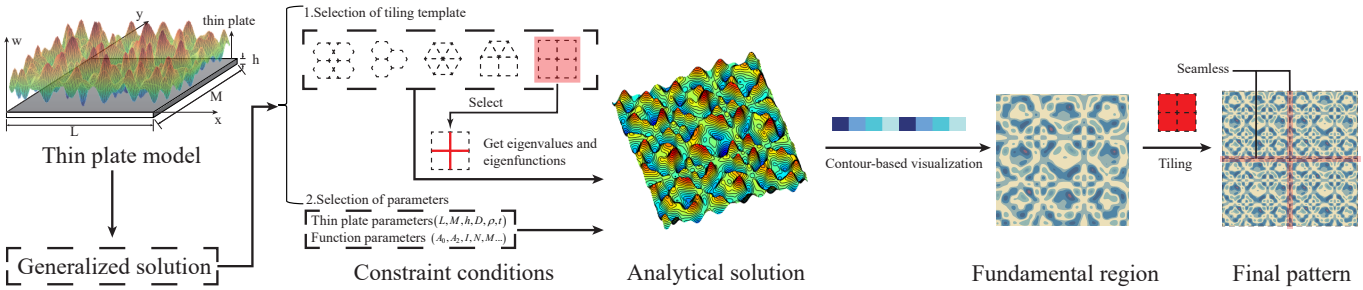

Fig. 1. Flowchart of the proposed aesthetic pattern generation pipeline.

and then constructing application-specific constraints to enforce symmetry, seamless tiling, and periodicity. Tiling-based approaches, in particular, rely on manually designed IMs within fundamental regions [1], [9], [12]. This raises design costs, restricts morphological flexibility, and may deform the fundamental region, harming aesthetic quality.

Inspired by Chladni plate experiments [4], [48], in which standing waves guide sand into regular nodal patterns (see Fig. 10 in [4] and Figs. 4.2–4.5 in [48]), we propose a PDE-based generative approach. Our goal is not to replicate physical Chladni patterns, but to leverage the conceptual framework of vibrational modes as a means of generating coherent aesthetic textures. By adjusting the parameters and boundary conditions of the PDE, each of which maps directly to a perceptible visual effect, the method generates rich patterns that naturally satisfy symmetry and seamless stitching constraints, obviating the need for explicit generation functions [7], [8], [26] or manually constructed IMs [1], [9], [12].

Thin-plate vibration is governed by the small-deflection bending equation [49], [50]:

$$D\nabla^4 w\left(x,y,t\right)+\rho h\frac{\partial^2 w\left(x,y,t\right)}{\partial t^2}=0. \tag{1}$$

Here,∇^4 is the biharmonic operator, and $D/(\rho h)$ determines the dispersion relation of the plane, where $D=Eh^3/[12(1-v^2)]$ is bending stiffness (E: elastic modulus, v: Poisson's ratio), ρ density, h thickness. The solution $w(x,y,t)$ represents transverse deflection. Following the visualization scheme of quasi-regular pattern [1], [7], [23], we apply color mapping to contours of w to generate diverse patterns.

Our pipeline (Fig.1) derives the general solution, defines boundary conditions according to the target tiling (symmetry and seamless constraints), solves for eigenvalues/eigenfunctions, and tiles the fundamental region across the canvas. This unified workflow integrates content generation and structural control, offering a highly interpretable and computationally efficient approach to aesthetic pattern creation.

IV. Equation Solution and Novel Boundary Condition Design

This section establishes the analytical foundation of our model. We derive the general solution of the thin-plate equation via separation of variables, obtaining trigonometric modes. Novel boundary conditions encoding symmetry and seamless tiling are then introduced to enable unified texture-layout control. Finally, we construct fundamental regions for both periodic and aperiodic tilings, linking the mathematical formulation to practical design.

A. General solutions

The free vibration equation of thin plates features parameters that directly influence the shape and scale variations in the contour lines of its solutions. Capitalizing on this property, we employ the Thin-Plate Free Vibration Equation (TPFVE) as the generative model. We first derive the general solution.

Applying separation of variables $w(x,y,t)=u(x,y)T(t)$ yields the following system of equations [50]:

$$\begin{cases} k^4=-\eta^4\dfrac{T''(t)}{T(t)}, \\ (\nabla^2+k^2)(\nabla^2-k^2)u(x,y)=0. \end{cases} \tag{2}$$

Where $\eta^2=\sqrt{\frac{\rho h}{D}}$. k is merely an intermediate variable introduced during the derivation, while $T(t)$ denotes the temporal component of the separated solution. This system decomposes into two independent PDEs: $(\nabla^2+k^2)u_1=0$ and $(\nabla^2-k^2)u_2=0$. The first is the Helmholtz equation, whose general solution is:

$$\begin{aligned} u_1(x,y)=\sum_{\alpha,\beta}&\left(A_0\cos(\alpha x)+A_1\sin(\alpha x)\right) \\ &\cdot\left(A_2\cos(\beta y)+A_3\sin(\beta y)\right), \end{aligned} \tag{3}$$

where α and β are the eigenvalues in the x- and y-directions arising from separation of variables applied to ∇^2. The coefficients A_i $(i=0,1,2,3)$ are undetermined. The second equation admits the solution:

$$\begin{aligned} u_2(x,y)=\sum_{\gamma,\delta}&\left(B_0\cosh(\gamma x)+B_1\sinh(\gamma x)\right) \\ &\cdot\left(B_2\cosh(\delta y)+B_3\sinh(\delta y)\right), \end{aligned} \tag{4}$$

with γ and δ being the corresponding eigenvalues, and B_i $(i=0,1,2,3)$ undetermined coefficients. Notably, the eigenvalues of the PDE are discrete and infinite in number; hence, u can be viewed as a superposition of distinct eigenfunctions. For Eq. (2), the following relationships and superposition principle hold:

$$\begin{cases} u=u_1+u_2, \\ \alpha^2+\beta^2=\gamma^2+\delta^2=k^2. \end{cases} \tag{5}$$

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TABLE I
THE EIGENVALUES AND EIGENFUNCTIONS FOR VARIOUS BOUNDARY CONDITIONS.

condition	Style	X/Y	Eigenvalues	Eigenfunctions
Neumann	free	X	$\alpha_0 = 0 \quad \alpha_n = \frac{n\pi}{L}, n = 1, 2, 3 \ldots$	$X_0(x) = A_0 \quad A_1 = 0, \quad X_n(x) = A_0 \cos(\alpha_n x)$
		Y	$\beta_0 = 0 \quad \beta_m = \frac{m\pi}{M}, m = 1, 2, 3 \ldots$	$Y_0(y) = A_2 \quad A_3 = 0, \quad Y_m(y) = A_2 \cos(\beta_m y)$
Dirichlet	(a) Bilateral symmetry(B_s)	X	$\alpha_0 = 0 \quad \alpha_n = f(n), n = 1, 2, 3 \ldots$	$X_0(x) = A_0 \quad X_n(x) = A_0 \cos(\alpha_n x) + A_1 \sin(\alpha_n x)$
		Y	$\beta_0 = 0 \quad \beta_m = \frac{2m\pi}{M}, m = 1, 2, 3 \ldots$	$Y_0(y) = A_2 \quad Y_m(y) = A_2 \cos(\beta_m y) + A_3 \sin(\beta_m y)$
	(b) Diagonal symmetry(D_s)	X	$\alpha_0 = 0 \quad \alpha_n = f(n), n = 1, 2, 3 \ldots$	$X_0(x) = A_0 \quad X_n(x) = A_0 \cos(\alpha_n x) + A_1 \sin(\alpha_n x)$
		Y	$\alpha_0 = 0 \quad \alpha_m = f(m), m = 1, 2, 3 \ldots$	$Y_0(y) = A_0 \quad Y_m(y) = A_0 \cos(\alpha_m y) + A_1 \sin(\alpha_m y)$
	(c) Isosceles triangle symmetry(I_s)	X	$\alpha_0 = 0 \quad \alpha_n = f(n), n = 1, 2, 3 \ldots$	$X_0(x) = A_0 \quad X_n(x) = A_0 \cos(\alpha_n x) + A_1 \sin(\alpha_n x)$
		Y	$\beta_0 = 0 \quad \beta_m = \frac{2(\varphi - m\pi)}{M}, m = 1, 2, 3 \ldots$ $\varphi = \arccos(\frac{A_2}{\sqrt{A_2^2 + A_3^2}})$	$Y_0(y) = A_2 \quad Y_m(y) = A_2 \cos(\beta_m y) + A_3 \sin(\beta_m y)$
	(d) Rotational symmetry(R_s)	X	$\alpha_0 = 0 \quad \alpha_n = \frac{2n\pi}{L}, n = 1, 2, 3 \ldots$	$X_0(x) = 0 \quad A_0 = 0 \quad X_n(x) = A_1 \sin(\alpha_n x)$
		Y	$\beta_0 = 0 \quad \beta_m = \frac{2m\pi}{M}, m = 1, 2, 3 \ldots$	$Y_0(y) = 0 \quad A_2 = 0 \quad Y_m(y) = A_3 \sin(\beta_m y)$

For the time-dependent ordinary differential equation (i.e., the second equation in Eq. (2)), its analytical solution can be derived straightforwardly and obtained:

$$T(t) = \left(I \cos\left(\frac{k^2}{\eta^2} t \right) + J \sin\left(\frac{k^2}{\eta^2} t \right) \right), \tag{6}$$

where I and J are undetermined coefficients. To simplify the solution, we set $B_i = 0$ $(i = 0, 1, 2, 3)$. This choice suppresses the hyperbolic (evanescent) wave components, which in a physical plate would only appear near free edges. The resulting pure trigonometric form is easier to control and visualize while preserving the essential oscillatory nature of vibration modes. This simplification does not compromise our generative objectives, as the lost flexibility is compensated by extensive boundary constraints and parameter tuning, which together provide sufficient flexibility for pattern generation. In the absence of boundary constraints, the final expression for $w(x, y, t)$ is obtained as:

$$\begin{aligned} w(x, y, t) = \sum_{\alpha, \beta} & \left(A_0 \cos(\alpha x) + A_1 \sin(\alpha x) \right) \\ & \cdot \left(A_2 \cos(\beta y) + A_3 \sin(\beta y) \right) \\ & \cdot \left(I \cos\left(\frac{\alpha^2 + \beta^2}{\eta^2} t \right) + J \sin\left(\frac{\alpha^2 + \beta^2}{\eta^2} t \right) \right). \end{aligned} \tag{7}$$

Since Eq. (7) admits an infinite number of particular solutions, it is essential to introduce boundary conditions to constrain the solution space and obtain discrete eigenvalues. This operation yields a series of discrete eigenvalues, denoted α_n and β_m.

B. Boundary conditions

The standard thin-plate equation is well established in mechanics; our contribution here is its systematic adaptation to pattern generation. We achieve this through three novel elements: (i) a classification of Dirichlet conditions encoding symmetries and tiling continuity, (ii) the concept of composite boundary conditions for multi-constraint enforcement, and (iii) closed-form eigenfunction solutions under these artistically motivated conditions.

The eigenvalues and eigenfunctions depend on the boundary conditions. In most thin-plate free vibration simulations, Neumann conditions (free edges) are adopted [48], as summarized in Tab. I and Fig. 2. To generate patterns with prescribed boundary geometries, we introduce Dirichlet conditions that enforce equality of function values on corresponding edges. This strategy offers two key benefits: it naturally imparts regular symmetry, and it directly extends to seamless tiling by enforcing continuity across tile edges.

Following this rationale, we apply Dirichlet conditions to derive tailored boundary schemes for the following four structures (Fig. 2):

(a) **Bilateral symmetry:** For a rectangular domain, equal function values are imposed on opposite edges (e.g., top and bottom):

$$w(l_1, 0, t) = w(l_1, M, t). \tag{8}$$

(b) **Diagonal symmetry:** Values at positions symmetric with respect to the rectangle's diagonal are set equal:

$$\begin{cases} w(l_1, 0, t) = w(0, l_1, t), \\ w(L, l_2, t) = w(l_2, M, t). \end{cases} \tag{9}$$

(c) **Isosceles triangular symmetry:** Two lines of opposite slope emanate from $(0, M/2)$ to the boundary, forming an isosceles triangular constraint that enforces symmetry about $y = M/2$:

$$w\left(l_1, k_0 l_1 + \frac{M}{2}, t \right) = w\left(l_1, -k_0 l_1 + \frac{M}{2}, t \right). \tag{10}$$

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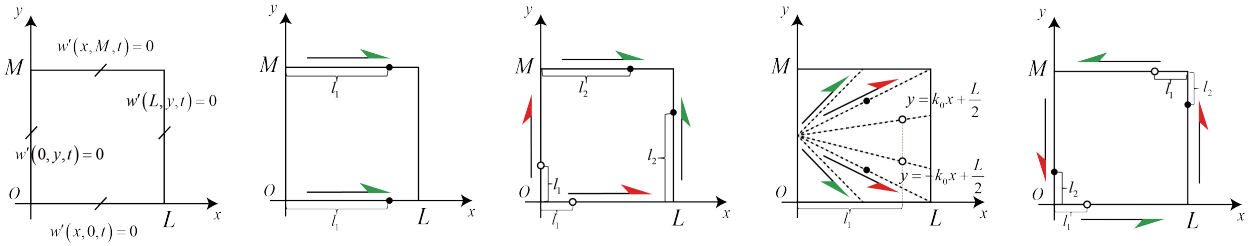

Fig. 2. Schematic of boundary conditions. Left: classical Neumann (free) condition. (a)–(d): four fundamental Dirichlet constraints proposed in this work. Arrows of the same color indicate corresponding edges; arrow directions indicate matching orientation; Hollow or solid points indicate that the function values under the corresponding constraints are equal.

(d) **Rotational symmetry:** Function values on the boundary are centrally symmetric about the plate center $(L/2, M/2)$:

$$\begin{cases} w(l_1, 0, t) = w(L - l_1, M, t), \\ w(0, l_2, t) = w(L, M - l_2, t). \end{cases} \tag{11}$$

The eigenvalues and eigenfunctions under these fundamental Dirichlet conditions are listed in Tab. I. Because these conditions constrain only part of the eigenvalue space, the remaining degrees of freedom can be expressed by an arbitrary function f. Consequently, a particular solution may simultaneously satisfy multiple boundary constraints, naturally leading to the concept of composite boundary conditions. Their general form is:

$$w(x, y, t) = \sum_{n=1}^{P} \sum_{m=1}^{Q} X_n(x)\, Y_m(y)\, T(t), \tag{12}$$

where for case (b) we require $P = Q$.

Practical tests show that single boundary constraints can produce patterns with expected symmetries, but they often do not meet diverse design requirements. Further analysis reveals that for certain constraints (e.g., B_s and I_s), strong restrictions apply mainly in one dimension; in others (e.g., D_s), the conditions only relate the boundaries without strictly limiting the eigenvalues. This flexibility allows a particular solution to satisfy multiple boundary conditions simultaneously, a scenario we term a **composite boundary condition**.

To further enrich the diversity of aesthetic patterns, we construct composite boundary conditions by superimposing multiple fundamental constraints. The procedure proceeds as follows: first, a fundamental boundary condition is selected from Tab. I and imposed along the x-dimension. If the y-dimension remains unconstrained, an additional boundary condition from Tab. I is imposed along y. Furthermore, when the pattern lacks symmetry about the line $y = x$, the diagonal symmetry condition D_s is additionally enforced. This yields six types of composite boundary constraints, as shown in Fig. 3 and illustrated in Fig. 4.

In total, ten distinct boundary conditions are formulated. The analytical solutions obtained under these constraints exhibit clearly defined symmetric structures, which in turn facilitate the systematic design of fundamental regions for various tilings, as demonstrated in Fig. 4.

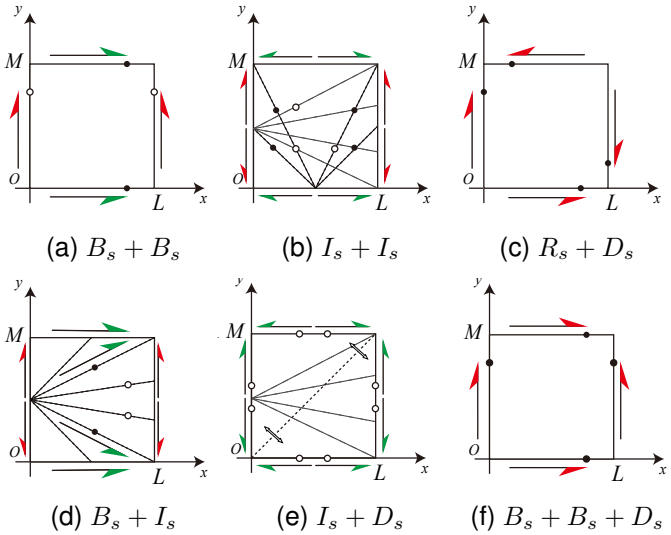

Fig. 3. Illustration of composite boundary conditions formed by superimposing basic Dirichlet constraints.

C. Solutions under tiling constraints

In contrast to patterns that focus predominantly on texture [6], [18], [23], tiling-based patterns (periodic or aperiodic) derive their value from the systematic repetition of coherent motifs, which is essential for seamless large-scale applications such as textiles, wallpapers, and architectural surfaces. This repetition provides a rhythmic foundation that accommodates a wide range of aesthetic intentions.

When generating tiling-based patterns, appropriate boundary constraints on tiles or fundamental regions are crucial to ensure seamless edge continuity and avoid artifacts. Most existing methods [1], [9], [12], [14], [40] rely on IMs, which can distort the fundamental region and make pattern quality sensitive to handcrafted functions [1].

To address this, we introduce a model integrating texture and layout based on the visualization scheme of QRP [1], [23]. Texture variation is achieved by tuning PDE physical parameters, while spatial layout is controlled via boundary conditions, eliminating complex IMs and ensuring intrinsic seamlessness.

We focus on widely used tiling structures to enhance controllability of spatial layout, systematically exploring compatible boundary conditions to establish both theoretical foundation and practical approach.

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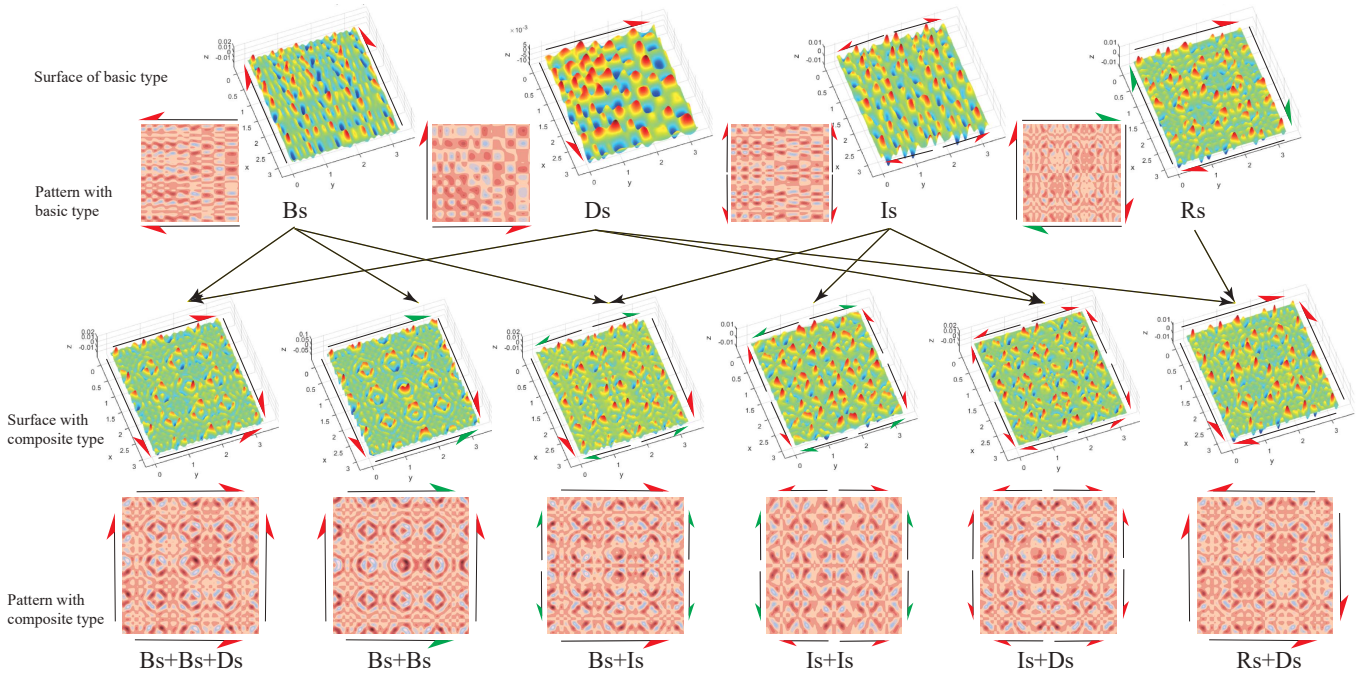

Fig. 4. Mapping from vibration modes to aesthetic patterns. Composite boundary conditions combine fundamental constraints to generate diverse symmetric patterns (see Fig. 3 for definitions of composite boundary conditions)

1) Periodic Tilings: Periodic tiling covers the plane by repeating a single tile without gaps or overlaps, using translation, rotation, or reflection. Unlike traditional methods requiring explicit IMs for each fundamental region, our PDE approach obtains solutions that inherently satisfy symmetry constraints by reformulating geometric conditions as boundary conditions.

Consider the k-uniform periodic tilings studied by Yin [1] (Fig. 5). The physical domain is the rectangle $a'b'c'd'$ ($-L/2 \le x' \le L/2$, $-M/2 \le y' \le M/2$), with colored subregions as fundamental regions. Continuity constraints on corresponding edges ensure seamless tiling. For dihedral symmetry, continuity is automatic; for rotational and reflectional symmetry, an I_s (isosceles triangular symmetry) condition suffices. After applying appropriate conditions, the three fundamental regions are periodically replicated across $a'b'c'd'$.

The physical domain is mapped to canvas $abcd$ ($0 \le x \le L$, $0 \le y \le M$) via:

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} S_x & 0 & L/2 \\ 0 & S_y & M/2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix}. \tag{13}$$

Where S_x, S_y are scaling factors that map the physical domain dimensions to the canvas coordinates.

Our model can also extend to periodic tilings of arbitrary quadrilaterals. Following Wang et al. [14] (quadrilaterals rotated about edge midpoints), we impose composite condition $I_s + I_s$ on the parametric rectangle, then map to the quadrilateral while preserving edge constraints. This guarantees functional continuity across adjacent tiles, ensuring seamlessness (Fig. 6), illustrating the flexibility of our PDE formulation.

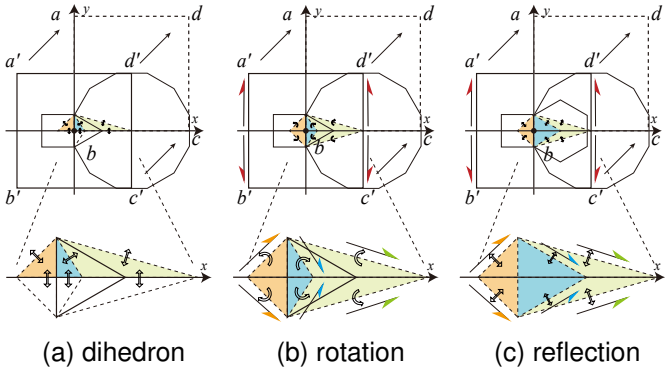

Fig. 5. Fundamental regions for three symmetry types in a periodic tiling: (a) dihedral, (b) rotational, (c) reflectional. Colored regions indicate fundamental regions: orange and green represent square and regular dodecagon; blue denotes an equilateral triangle in (a,b) and a regular hexagon in (c). Edges e and e' connected by same-color arrows satisfy $h(e) = h(e')$.

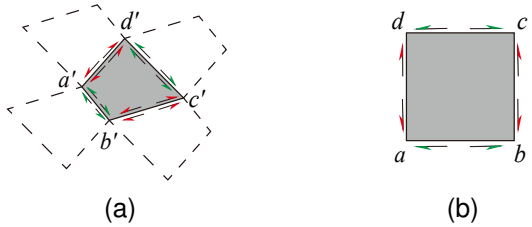

Fig. 6. Mapping between the fundamental region $a'b'c'd'$ and the parametric domain $abcd$ for quadrilateral tilings.

2) Aperiodic Tilings: Aperiodic tilings lack translational symmetry, giving rise to greater structural diversity and more intricate boundary conditions than periodic tilings. This complexity makes it difficult to precisely define fundamental

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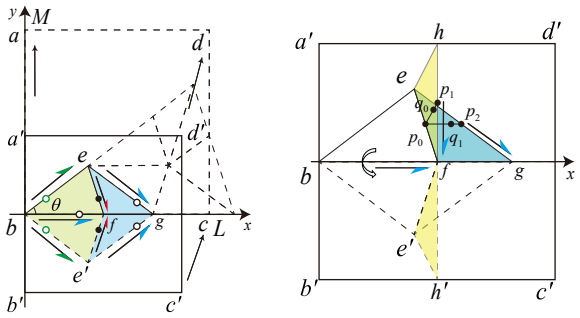
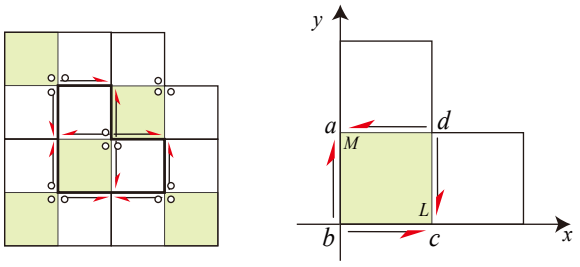


(a) Fundamental region (b) Affine mapping

Fig. 7. Fundamental regions of the Robinson triangle tiling. (a) $\triangle bfe$ and $\triangle fge$ with three constraints: $h(ef) = h(e'f)$, $h(be) = h(be')$, $h(bf) = h(eg) = h(e'g)$. (b) Affine transformation mapping $\triangle efh$ onto $\triangle efg$ to satisfy $h(bf) = h(hf) = h(eg)$. Here, the scalars $L = bc$ and $M = ab$ denote the dimensions of the canvas $abcd$ along the two coordinate directions, respectively.



(a) Fundamental region (b) basic edge conditions

Fig. 8. Fundamental regions of the chair tiling. The pattern function must satisfy $h(da) = h(ba) = h(bc) = h(dc)$. L and M have the same meaning as in Fig. 7.

regions and construct IMs. We investigate two representative cases: Robinson triangles, a well-known Penrose-tiling variant (collectively referred to as "Penrose tilings" per the convention in [12]), and chair tilings [39]. These examples demonstrate how shape-matching, adjacency, and local symmetry compensation can be realized through constrained boundary conditions.

Robinson triangle tiling consists of two kinds of isosceles triangles. Fig. 7(a) illustrates its boundary constraints. Based on the aperiodic structure, fundamental regions are arranged over physical domain $a'b'c'd'$. Interior points (x', y') are mapped to canvas $abcd$ via an affine transformation similar to periodic tiling:

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} S_x & 0 & 0 \\ 0 & S_y & S_y\tau \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix}, \tag{14}$$

where the rectangle has side length 2τ ($\theta = \pi/5$) and centroid $f = (\tau, 0)$. Under composite condition $I_s + D_s$, conditions 1 and 2 in Fig. 7 hold, but condition 3 does not. Since $I_s + D_s$ ensures $h(bf) = h(hf)$, we apply an affine mapping of $\triangle efh$ onto itself (Fig. 7(b)).

Let $q_1 = (x', y')$ be interior to $\triangle efg$. Intersect $y = y'$ with edges ef and eg at p_0 and p_2. Map p_2 proportionally to p_1 on hf:

$$\frac{\|hp_1\|}{\|hf\|} = \frac{\|ep_2\|}{\|eg\|}. \tag{15}$$

Next, take a point q_0 on the segment p_0p_1 that satisfies

$$\frac{\|q_0p_1\|}{\|p_0p_1\|} = \frac{\|q_1p_2\|}{\|p_0p_2\|}, \tag{16}$$

with p_0, p_1, and p_2 given by

$$\begin{cases} p_0 = \big((\tau + \tau\sin\theta - y')\cot 2\theta + \tau\cos\theta,\ y'\big), \\ p_1 = \big(\tau,\ \tau - (\tau - y')\csc\theta\big), \\ p_2 = \big((\tau + \tau\sin\theta - y')\cot\theta + \tau\cos\theta,\ y'\big). \end{cases} \tag{17}$$

For any interior point $q_0 \in \triangle efg$, the corresponding point $q_1 \in \triangle efh$ is obtained via the mapping:

$$\begin{cases} q_0 = \lambda p_0 + (1-\lambda)p_1, \\ \lambda = \dfrac{p_{2x} - x'}{p_{2x} - p_{0x}}. \end{cases} \tag{18}$$

Chair tiling assembles three squares into a "chair-shaped" tile (Fig. 8). The relation $h(da) = h(ba) = h(bc) = h(dc)$ holds under composite condition $R_s + D_s$. Using this, we generate rectangular tile $abcd$ and distribute it according to chair tiling rules.

However, $R_s + D_s$ makes $abcd$ symmetric about ac, potentially causing monotony. To counter this, we add a perturbation $g(x, y)$ that vanishes at boundaries:

$$\begin{cases} w(x,y,t) = \displaystyle\sum_{n=1}^{P}\sum_{m=1}^{Q} X_n(x)Y_m(y)T(t) + g(x,y), \\ g(x,y) = x(L-x)\,y(M-y)\,\sin(\omega_0 x)\cos(\omega_1 y). \end{cases} \tag{19}$$

Since g vanishes at the fundamental region's boundary, the tile remains seamless while breaking symmetry and enriching visual variety.

V. Rendering of Patterns

For a fixed time t, the analytical solution $w(x, y, t)$ is treated as a height field $u(x, y)$. To generate a visual pattern, we map the range of u onto a finite set of color intervals. Specifically, the range is divided into R user-defined intervals $h_1, h_2, \ldots, h_R$. A color palette assigns a distinct color to each interval. The final pattern is produced by coloring each pixel according to the interval into which the height value at its corresponding sample point falls.

The rendering process is fully executed on the GPU to achieve high efficiency. The analytical expression of $w(x, y, t)$, along with its parameters, the threshold values $h_1, \ldots, h_R$, and the color palette, are passed to shaders as uniform variables. During rasterization, the vertex shader passes geometric and texture coordinates to the fragment shader. For each fragment, the fragment shader evaluates $w(x, y, t)$ at the fragment's position, determines the interval containing the resulting height via a fast threshold search (e.g., a binary or linear scan over the sorted thresholds), and outputs the corresponding color.

This GPU-based strategy exploits the massive parallelism of modern graphics hardware, enabling simultaneous processing

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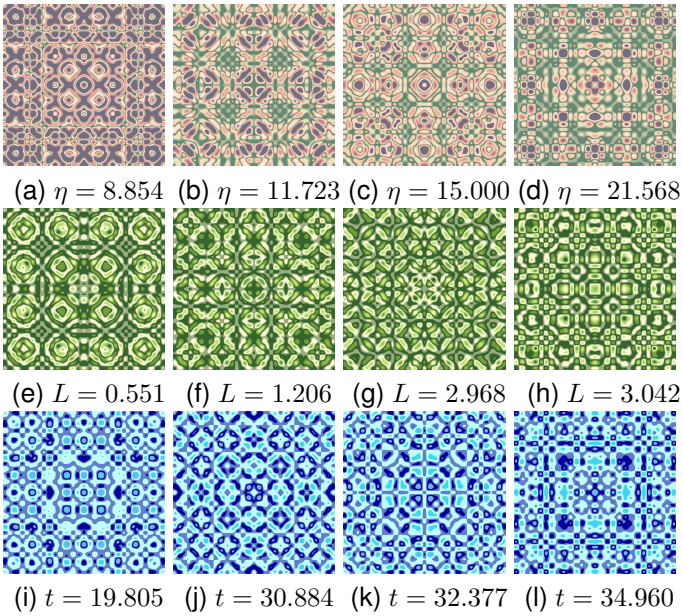

Fig. 9. Effect of strong parameters η, $L(M)$, and t under $I_s + D_s$ (with $L = M$ by default). Parameter sets: (a)-(d) $\{23.059, \eta, 1.781, 44.064, 16, 16, 1.53, 1.53\}$; (e)-(h) $\{31.279, 15, L, 4, 18, 18, 0.434, 0.434\}$; (i)-(l) $\{t, 15, 3.236, 4, 18, 18, 0.434, 0.434\}$.

of millions of fragments. As reported in Sec. VI, our implementation renders patterns at resolutions up to 4096×4096 in approximately 0.33 ms on a consumer GPU, thus supporting interactive exploration of design parameters. The complete rendering pipeline can be summarized as:

1) **Data preparation:** The thresholds $(h_1, \ldots, h_R)$, color palette (RGB or RGBA values), tile geometry, and PDE parameters $(\eta, L, M, P, Q, A_0, A_2, t,$ etc.) are uploaded to the GPU as uniform variables.
2) **Vertex shader:** It transforms tile vertices from object space to screen space and forwards texture coordinates to the fragment shader.
3) **Fragment shader:** It computes $w(x, y)$ for the current fragment using the analytical formula (evaluated in parallel), performs a threshold search to select the appropriate color, and outputs the colored fragment.
4) **Framebuffer output:** The final colored fragments are written to the framebuffer and displayed.

This parallel approach reduces rendering time dramatically compared to CPU-based serial computation, enabling real-time parameter tuning and interactive design workflows.

VI. Experimental Results

The proposed pattern generation algorithms were implemented in C++ and integrated with the OpenGL graphics pipeline for efficient rendering. Core numerical evaluations are offloaded to the GPU, where per-pixel function values are computed in parallel using GLSL shaders. All experiments were conducted on a workstation equipped with an Intel Core i5-12600K CPU, an NVIDIA GeForce RTX 3060 Ti GPU, and the Windows 10 operating system.

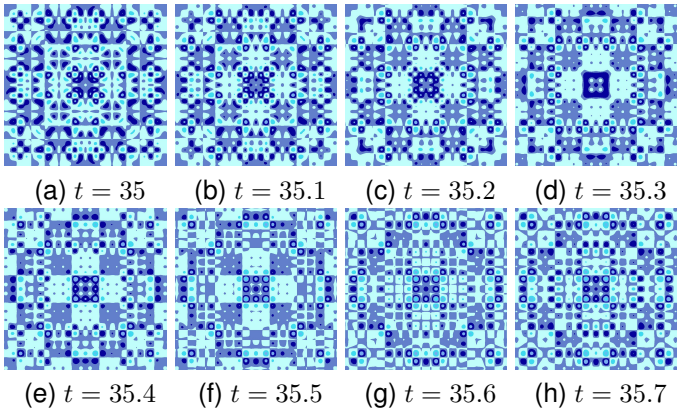

Fig. 10. Dynamic pattern evolution driven by time t under $I_s + D_s$, with parameters $\{t, 15.883, 3.236, 4, 16, 16, 0.434, 0.434\}$.

We conducted a series of controlled experiments to examine the influence of physical parameters, boundary conditions, and tiling structures. The algorithm synthesizes varied patterns through three primary mechanisms: (1) tuning key physical parameters, (2) selecting different symmetry modes, and (3) applying various tiling layouts (periodic and aperiodic). To ensure consistent boundary conditions, we set $L = M$ throughout all experiments. For notational simplicity, A_1 and A_3 are fixed to 1 unless otherwise specified. The core parameter set considered is $\{t, \eta, L, I, P, Q, A_0, A_2\}$. For reproducibility, the full parameter configurations for all generated patterns are provided in the corresponding figure captions.

A. Parameter Control

The model parameters, though derived from the plate equation, directly map to perceptible visual qualities. We distinguish two categories: *strong parameters*, which control global texture structure and morphology, and *weak parameters*, which refine local details and stylistic expression.

1) Strong Parameters (Global Morphology): The following three parameters determine the overall appearance of the pattern.

The composite parameter η. Defined as $\eta^2 = \sqrt{\rho h / D}$, this parameter consolidates the physical quantities mass density ρ, plate thickness h, and bending stiffness D into a single control. Visually, η governs the overall texture morphology: different values yield distinctly different styles of patterns (Figs. 9(a)–(d)).

The plate dimensions $L(M)$. These parameters specify the physical length and width of the plate, respectively, and thus determine the size of the solution domain. When the solution is mapped onto a fixed canvas, larger $L(M)$ yields a wider spatial extent of the vibration pattern, effectively displaying more texture cycles within the same frame. Conversely, smaller $L(M)$ confines the pattern to a more localized region, resulting in fewer visible cycles but richer, more finely resolved details (Figs. 9(e)–(h)).

The time parameter t. The solution contains a time-harmonic term; varying t continuously produces smooth, wave-like motion (Fig. 10). The pattern undergoes periodic deformation with a period determined by $\eta^2/(\alpha^2 + \beta^2)$,

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enabling temporal animation and live preview. This temporal degree of freedom is unique to our physics-based formulation and offers a dynamic dimension not present in static texture generators.

2) Weak Parameters (Local Refinement): The following parameters fine-tune the internal texture without altering the global structure.

Coefficients A_0 and A_2. These control directional elongation. Increasing A_0 stretches the pattern along the x-axis; increasing A_2 stretches it along the y-axis (Figs. 11(a)–(h)). This allows independent control of horizontal and vertical texture anisotropy.

Coefficient I. This regulates textural hierarchy by controlling the amplitude of the time-dependent oscillation. Higher I enhances layering and depth, producing more three-dimensional, intricate details (Figs. 11(i)–(l)).

Summation limits P and Q. These determine the number of eigenmodes included in the superposition. Larger P, Q introduce higher spatial frequencies, resulting in denser, more complex textures (Figs. 11(m)–(p)). The computational cost scales linearly with $P \times Q$, offering a trade-off between detail and performance.

Additive perturbation (Eq. (19)). A boundary-vanishing perturbation $g(x, y)$ can be superimposed to break symmetry, converting symmetric patterns into asymmetric variants without modifying the boundary conditions (Fig. 12). This provides additional stylistic flexibility beyond the core parameter space.

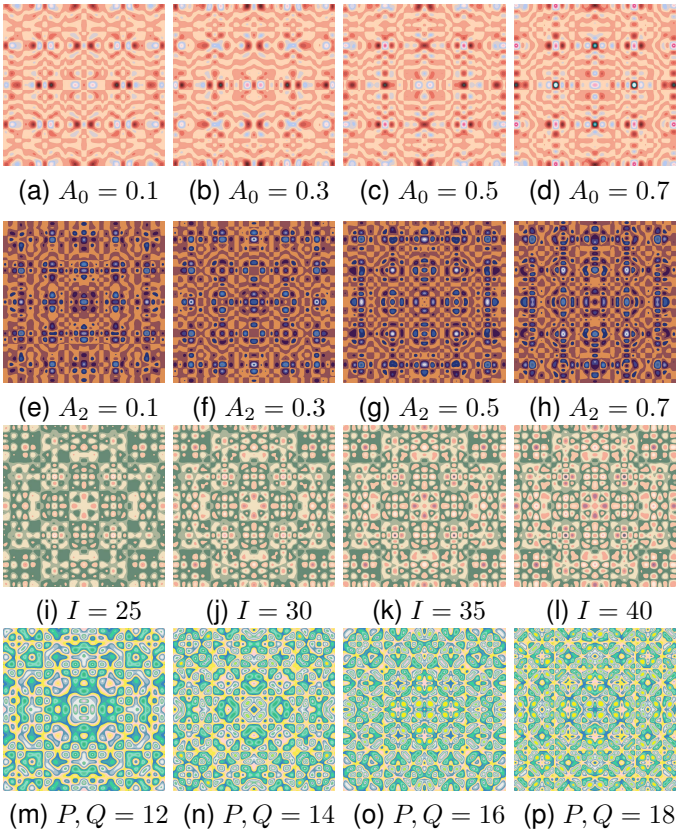

Fig. 11. Effect of weak parameters A_0, A_2, I, P, Q under the $I_s + D_s$ boundary condition. Parameter sets used: (a)–(d) $\{11.187, 7.436, 3.326, 14.384, 16, 16, A_0, 1.256\}$; (e)–(h) $\{36, 15, 3.326, 4, 18, 18, 0.434, A_2\}$; (i)–(l) $\{16.438, 18.735, 1.804, I, 16, 16, 1.2, 1.2\}$; (m)–(p) $\{30.941, 15, 3.236, 38.691, P, Q, 0.434, 0.434\}$.

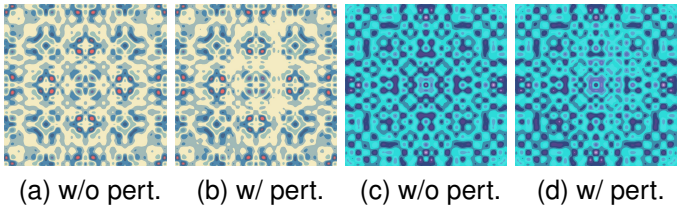

Fig. 12. Effect of the additive perturbation term on patterns generated under $I_s + D_s$. Parameter sets: (a)-(b) $\{24.8, 16.284, 2.157, 4, 16, 16, 0.434, 0.434\}$; (c)-(d) $\{28.971, 12.348, 3.236, 7.764, 16, 16, 0.434, 0.434\}$.

TABLE II
SUMMARY OF MODEL PARAMETERS AND THEIR VISUAL EFFECTS.

Parameter	Type	Visual Effect
η	Strong	overall appearance / style
$L(M)$	Strong	Spatial extent / texture cycles / details
t	Strong	Pattern animation / phase shift
A_0, A_2	Weak	Directional elongation
I	Weak	Hierarchical depth / layering
P, Q	Weak	Texture complexity / detail
$g(x, y)$	Weak	Symmetry breaking

For reproducibility, the full parameter configurations for all generated patterns are provided in the corresponding figure captions. Fig. 9 demonstrates the effect of strong parameters, and Fig. 11 illustrates the effect of weak parameters under fixed $I_s + D_s$ conditions.

Users need not understand the underlying mechanics to use these parameters effectively; each control is presented in terms of its direct visual consequence. Moreover, the parameter-to-output mapping is continuous and predictable: small adjustments produce gradual, perceptible changes, which aligns well with the exploratory trial-and-error workflow common in artistic practice. This stands in contrast to chaotic systems, where small parameter changes can lead to unpredictable jumps in output. A summary of all parameters and their visual effects is provided in Tab. II.

B. Boundary Condition Control

With $\{t, \eta, L, I, P, Q\}$ fixed, the coefficients A_0 and A_2 determine the resulting composite boundary conditions. Fig. 13 presents a set of aesthetic patterns generated under different composite boundary conditions. Each combination produces distinct symmetries:

- $B_s + B_s$ (a): bilateral symmetry in both directions, yielding a four-fold mirrored pattern.
- $I_s + I_s$ (b): isosceles triangular symmetry in both axes, producing intricate diamond-like motifs.
- $R_s + D_s$ (c): rotational plus diagonal symmetry, generating spiral-like structures.
- $B_s + I_s$ (d): mixed bilateral and triangular symmetry, creating asymmetric but balanced layouts.
- $I_s + D_s$ (e): triangular-diagonal combination, giving complex star-shaped patterns.

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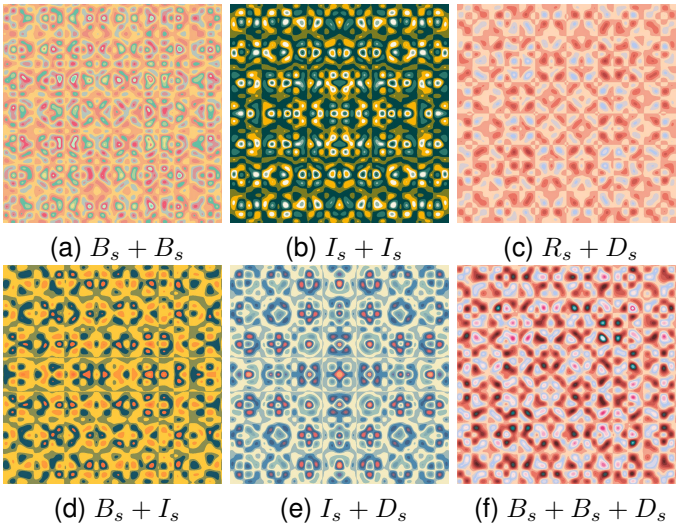

Fig. 13. Aesthetic patterns generated under different composite boundary conditions with fixed parameters. Parameter sets: (a) $\{26.256, 15, 3.236, 4, 18, 18, 0.214, 2.363\}$; (b) $\{26.256, 15, 3.236, 4, 18, 18, 0, 0.685\}$; (c) $\{26.256, 15, 3.236, 4, 18, 18, 0, 0\}$; (d)–(f) $\{26.256, 15, 3.236, 4, 18, 18, 0.434, 0.434\}$.

- $B_s + B_s + D_s$ (f): triple composite, the most constrained, resulting in highly regular, almost crystalline textures.

Users can select suitable composite conditions according to specific design needs, achieving precise control over symmetry characteristics.

Notably, the patterns in Figs. 13(a), (b), (d), (e) and (f) inherently possess strong tiling properties due to their underlying mathematical structure. This built-in seamless repeatability is a key advantage of our method, providing a practical foundation for applications requiring continuous visual composition, such as textile design and decorative art.

C. Tiling Structure Control

Tiling structures provide an efficient framework for generating complex patterns with rich hierarchical details. We evaluate periodic and aperiodic tilings on several classical examples.

Periodic tilings. Figs. 14 and 15 show results for k-uniform and quadrilateral-based periodic tilings. Even within a fixed tiling structure, varying physical parameters and boundary conditions yields distinctly different stylistic outcomes. Under the I_s boundary condition, k-uniform tilings can maintain seamless boundaries while allowing independent adjustment of the eigenvalue in one dimension, thereby producing different pattern variations and generating diverse visual effects (Fig. 14). The quadrilateral tilings (convex and concave) demonstrate that our method adapts to non-rectangular fundamental regions without distortion artifacts, preserving texture coherence across tile boundaries. Under the $I_s + I_s$ condition, the method is able to generate textures consistently across these tilings. This parametric flexibility underscores the creative potential of periodic tilings (Fig. 15).

Aperiodic tilings. Robinson triangle (Penrose-type) and chair tilings (Fig. 16) lack translational symmetry but preserve long-range order. This inherent symmetry-breaking introduces controlled, organic variability, mitigating the visual monotony often associated with purely periodic systems. For Penrose-type tilings, the $I_s + D_s$ condition generates smoothly varying textures across dart and kite tiles (Fig. 16(b–d)). For chair tilings, $R_s + D_s$ produces radial patterns that maintain continuity at tile boundaries (Fig. 16(f–h)). Our PDE-based model adapts to these complex layouts through tailored boundary conditions, generating patterns that are ordered yet never repetitive.

Design workflow. A practical design workflow emerges from the flexibility of our framework: first choose a suitable tiling template, then adjust strong parameters to establish core texture and global morphology, and finally fine-tune weak parameters to refine local details and stylistic expression. As shown in Fig. 17, selecting different tiling templates and parameter configurations yields a rich variety of stylistic outcomes. The composite boundary conditions eliminate the need for hand-crafted invariant mappings while avoiding excessive homogeneity, establishing a unified, efficient, and highly controllable foundation for tiling-based aesthetic pattern creation.

D. Comparisons

The preceding subsections (Sec.VI-A to VI-C) have demonstrated the controllability of our model, illustrating how pattern morphology, symmetry, and tiling layout can be intuitively steered via physical parameters, boundary conditions, and tiling schemes. To further validate its practical advantages, we present a direct comparative study with conventional methods, focusing on output quality, controllability, and computational flexibility. The results provide empirical evidence that our unified PDE-based model not only simplifies the generation workflow but also enhances expressive capability and operational ease.

Texture richness and diversity.

Figure 18 compares patterns generated from quasi-regular functions [7], [23], dynamical systems [29], and our Thin-Plate Free Vibration Equation (TPFVE). Patterns produced by the TPFVE exhibit two distinct advantages: (i) highly varied yet coherent composition, integrating multiple visually harmonious elements within a single pattern; (ii) intuitive and structured control, where pattern shapes are effectively tuned via physical parameters and boundary conditions, avoiding the complex explicit function design typical of dynamical systems [6].

Invariant mappings vs. boundary conditions. Traditional seamless tiling is often achieved by constructing IMs over predefined tiling units. While ensuring continuity, this approach frequently leads to limited visual diversity, perceptible discontinuities, and patterns that deviate from the intended style of the base motif. Figure 19 directly compares IM $(x = \sin(2\pi x/L), y = \cos(2\pi y/M + \pi))$ with our boundary condition method on a square tiling. Textures generated by IM exhibit noticeable homogeneity and structural repetition, whereas our method produces substantially richer and more organic detail. This improvement stems from the inherent properties of the B_s and I_s conditions: B_s reduces symmetry

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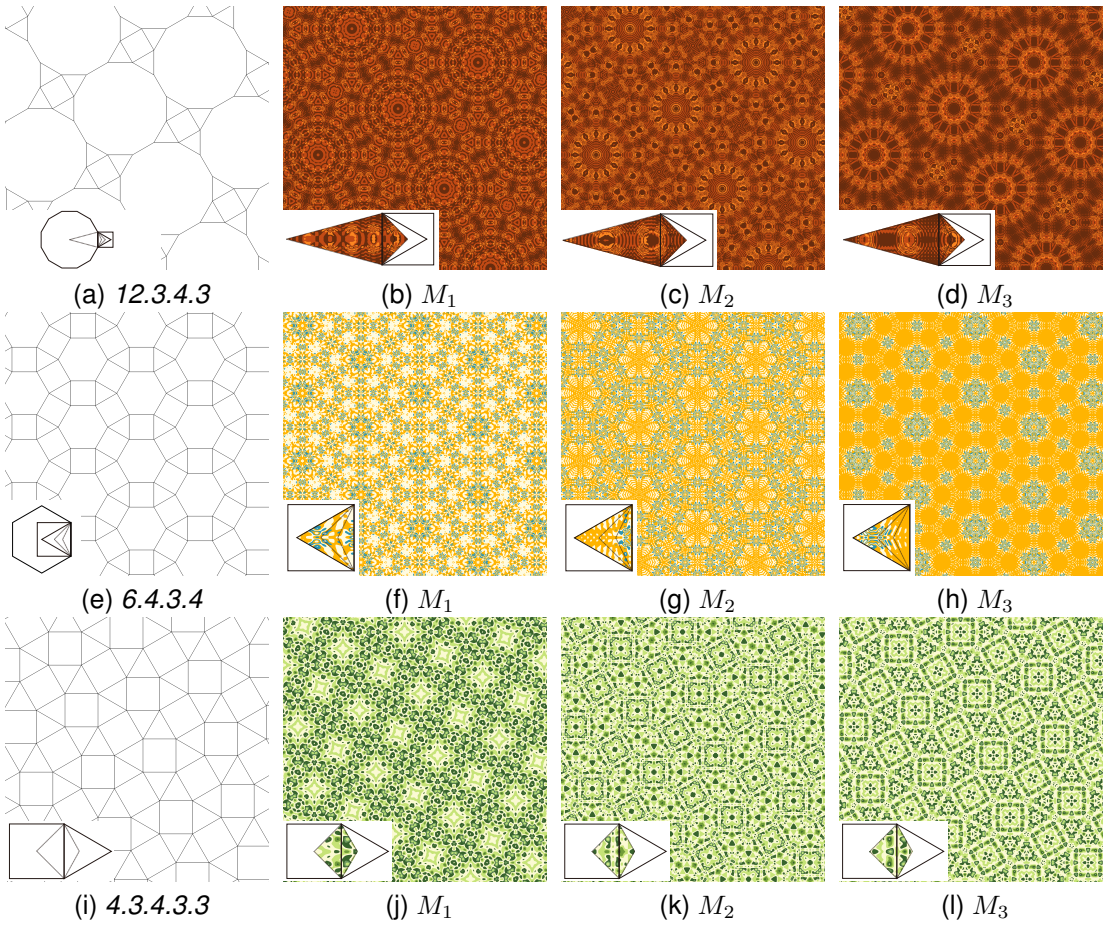

Fig. 14. Aesthetic patterns generated under different periodic tilings, shown with their fundamental regions (insets in the lower-left corners). (a), (e), (i) Archimedean tiling templates. α can be defined as an arbitrary function. We set α as follows: $M_1 : \alpha_n = n\pi/L$; $M_2 : \alpha_n = 2(\varphi - n\pi)/L$; $M_3 : \alpha_n = (2n+1)\pi/L$. (b)–(d) 12.3.4.3 structure with parameters $\{18.7, 12.7, 1.7, 2.5, 27, 27, 5.9, 8.4\}$. (f)–(h) 6.4.3.4 structure with $\{15.6, 3.4, 2.2, 2.5, 24, 24, 2.9, 5.2\}$. (j)–(l) 4.3.4.3.3 structure with $\{9.9, 15.4, 2.2, 2.5, 14, 14, 1, 8\}$. All periodic tilings use the I_s boundary condition ($\beta_m = 2(\varphi - m\pi)/M$).

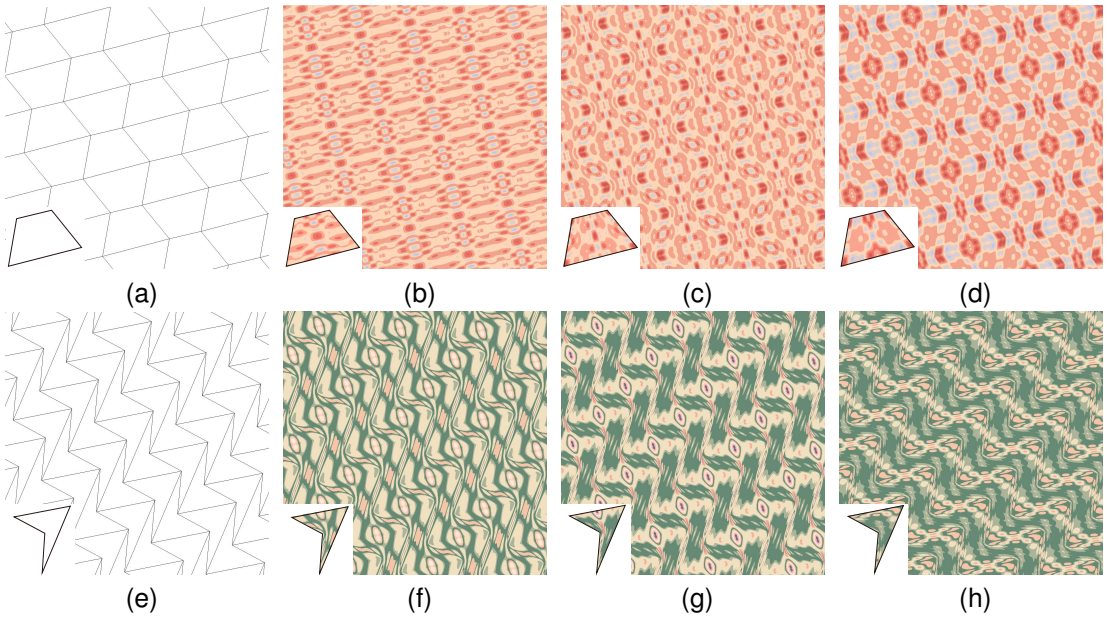

Fig. 15. Aesthetic patterns under convex and concave quadrilateral tilings using $I_s + I_s$ boundary condition, with insets of the fundamental quadrilaterals in the lower-left corners. (a) and (e) show tiling templates. (b)–(d) convex: parameters (b) $\{4.216, 18.536, 1, 15.723, 3, 6, 0.745, 0.434\}$, (c) $\{19.314, 7.37, 1, 17.157, 7, 3, 0.422, 0.931\}$, (d) $\{36.765, 13.375, 1, 3.627, 5, 2, 0.422, 0.931\}$. (f)–(h) concave: (f) $\{8.235, 12.414, 1, 54.902, 4, 2, 0.833, 1.167\}$, (g) $\{10.49, 8.186, 1, 121.471, 5, 2, 2.931, 0.598\}$, (h) $\{33.824, 14.431, 1, 40.49, 2, 7, 0.434, 0.434\}$.

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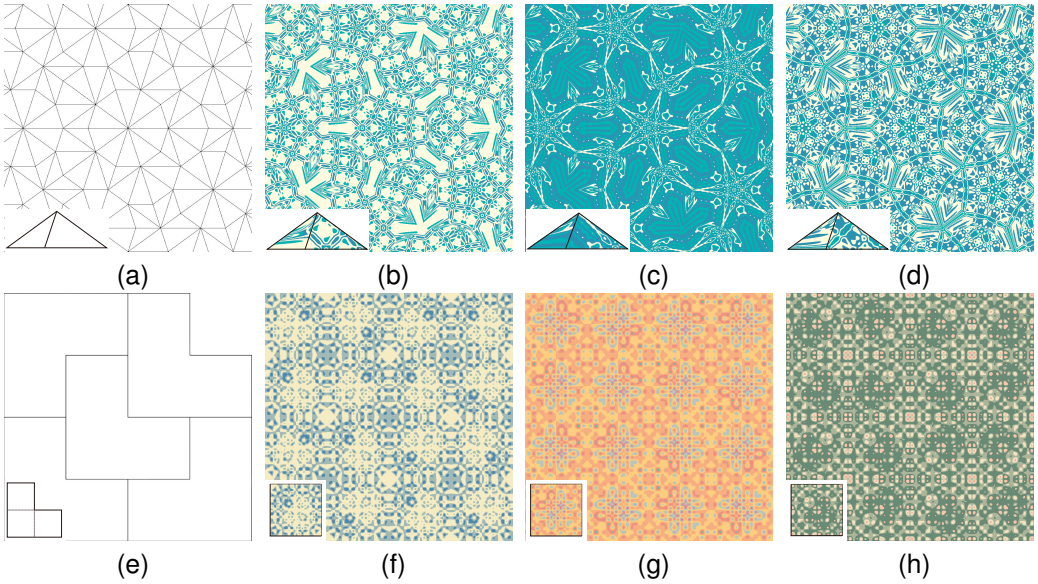

Fig. 16. Aesthetic patterns under aperiodic tilings, with insets of the fundamental regions in the lower-left corners. (a) Robinson triangle tiling template (a Penrose-type aperiodic tiling); (b)–(d) patterns under $I_s + D_s$ with parameters (b) $\{3.4, 5, 3.236, 2.5, 11, 11, 1, 1\}$, (c) $\{9.8, 21.3, 3.236, 2.5, 11, 11, 1, 1\}$, (d) $\{18.2, 12, 3.236, 2.5, 11, 11, 1, 1\}$. (e) Chair tiling template; (f)–(h) patterns under $R_s + D_s$ with parameters (f) $\{18.037, 15.49, 1.902, 9.978, 11, 11, 0, 0\}$, (g) $\{36.184, 2.085, 1.749, 21.107, 9, 9, 0, 0\}$, (h) $\{30.757, 12.884, 3.015, 69.846, 9, 9, 0, 0\}$.

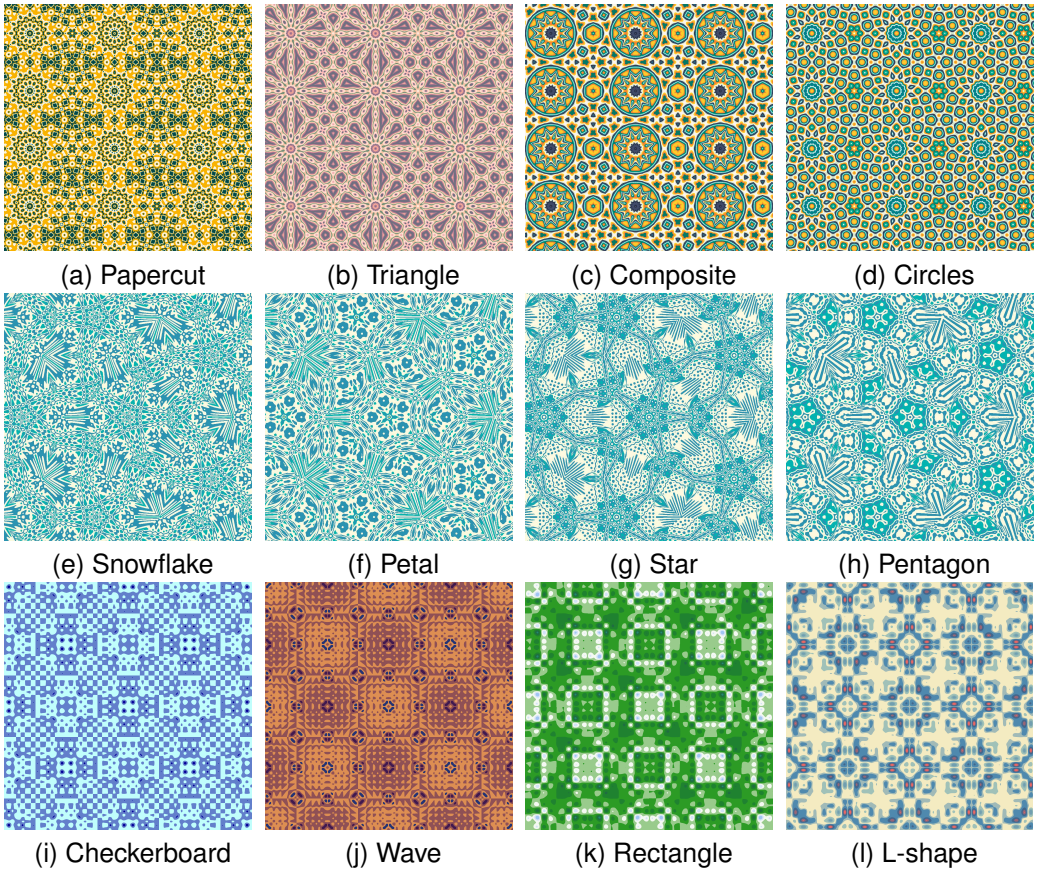

Fig. 17. Diverse aesthetic patterns generated under different tiling structures and parameter configurations. (a)–(d): [33434; 334.12; 3464; 3.12.12; 46.12] M_2 periodic tiling with parameters (a) $\{75, 12, 2, 10, 8, 8, 5, 8\}$, (b) $\{41, 59, 6, 10, 8, 8, 5, 8\}$, (c) $\{17, 7, 9, 10, 8, 8, 5, 8\}$, (d) $\{108, 63, 8, 10, 8, 8, 5, 8\}$. (e)–(h) Robinson tiling with (e) $\{10, 25, 3.236, 2.5, 8, 8, 1, 1\}$, (f) $\{16, 14, 3.236, 2.5, 8, 8, 1, 1\}$, (g) $\{7, 16, 3.236, 2.5, 8, 8, 1, 1\}$, (h) $\{14, 18, 3.236, 2.5, 8, 8, 1, 1\}$. (i)–(l) Chair tiling with (i) $\{25.210, 13.333, 1.457, 4, 10, 10, 0, 0\}$, (j) $\{14.706, 7.637, 1.457, 3.922, 10, 10, 0, 0\}$, (k) $\{5.602, 14.157, 0.490, 22.549, 8, 8, 0, 0\}$, (l) $\{18.067, 9.559, 1.667, 49.02, 6, 6, 0, 0\}$.

to encourage diversity, while I_s preserves full symmetry for uniform and coherent outcomes. Users may select the appropriate condition based on their visual preference.

Periodic and aperiodic tilings. We further compare our

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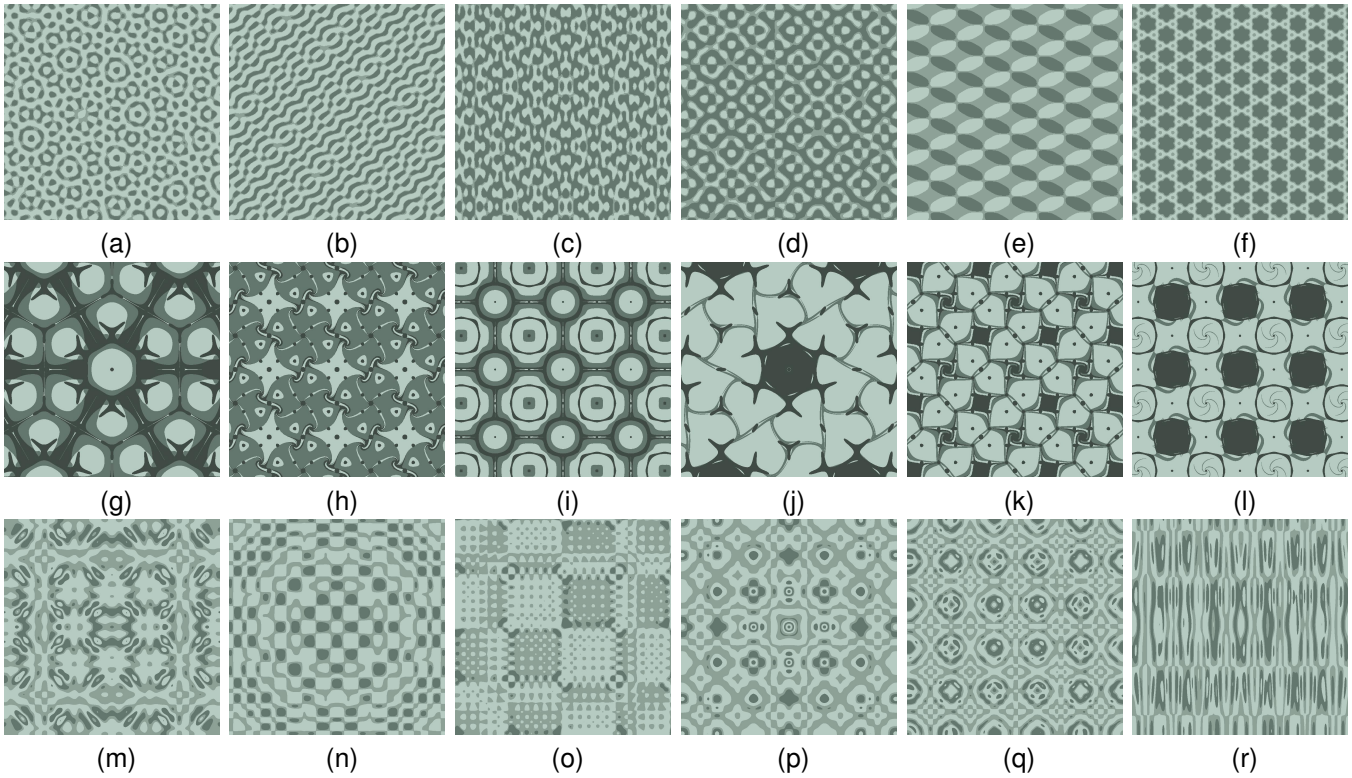

Fig. 18. Comparison of texture patterns generated by (a)–(f) QRP, (g)–(l) DS, and (m)–(r) our TPFVE.

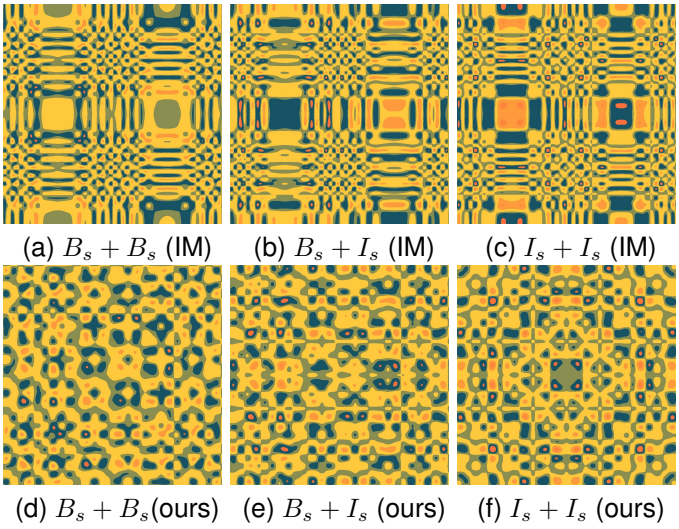

Fig. 19. Comparison between traditional invariant mapping (IM) and our boundary conditions for square tilings. Parameters $\{25.358, 15, 3.109, 7, 16, 16, 0.434, 0.434\}$. (a,d) $B_s + B_s$; (b,e) $B_s + I_s$; (c,f) $I_s + I_s$. IM uses $x = \sin(2\pi x/L)$, $y = \cos(2\pi y/M + \pi)$.

method against invariant mapping (IM) on both k-uniform periodic tilings [1] and Robinson triangle tilings (a Penrose-type aperiodic family) [12], as shown in Fig. 20. For a fair comparison, we maintain identical TPFVE parameters across methods: in Figs. 20(a)–(f), textures are generated using the same TPFVE model but tiled via either IM or our boundary conditions; in Figs. 20(g)–(i), textures are produced by combining IM with either a QRP function or the TPFVE solution. Patterns tiled with IM appear monotonous and repetitive, while those generated under our boundary conditions are significantly richer and more varied. This divergence reflects a fundamental difference in generative philosophy: IM defines explicit functional correspondences between tile edges, inherently restricting the solution space; our method encodes tiling constraints as PDE boundary conditions, allowing the governing equation to satisfy continuity and symmetry organically and supporting a broader spectrum of intrinsic solutions.

Arbitrary quadrilateral tilings. We extend the comparison to arbitrary quadrilateral tilings following the method of Wang et al. [14], as shown in Fig. 21. Textures generated via IM combined with either QRP or DS models often exhibit noticeable local distortion. This artifact arises because mapping a regular parametric domain to a highly deformed quadrilateral unevenly stretches or compresses texture features. In contrast, our method adapts naturally to the tile geometry, preserving textural coherence and detail without manual adjustment. This capability highlights a key practical advantage: our model maintains visual fidelity across a wide variety of tile shapes, reducing the need for case-by-case manual tuning and supporting more robust design workflows.

Collectively, these comparisons demonstrate that our unified PDE-based approach not only outperforms conventional methods in terms of visual diversity and structural coherence but also offers a more intuitive and flexible control mechanism for aesthetic pattern generation.

E. Applications

The aesthetic patterns generated by our method find broad utility across both creative and technical domains. In creative

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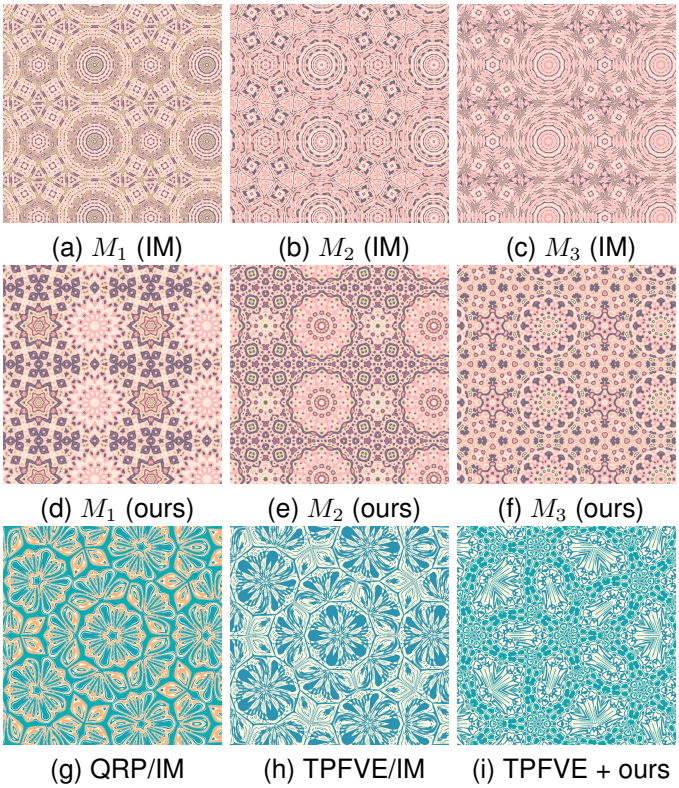

Fig. 20. Comparison of our method with invariant mapping (IM) on k-uniform periodic tilings [1] and Robinson triangle aperiodic tilings [12]. (a)–(f): k-uniform tilings [33434; 334.12; 3464; 3.12.12; 46.12] with parameters $\{6.4, 6.6, 4.5, 2.5, 16, 16, 1, 12.4\}$ and eigenvalue formulations $M_1 : \alpha_n = n\pi/L$, $M_2 : \alpha_n = 2(\varphi - n\pi)/L$, $M_3 : \alpha_n = (2n + 1)\pi/L$, with $\beta_m = 2(\varphi - m\pi)/M$. (g)–(i): Robinson triangle tilings: (g) QRP with IM, (h) TPFVE with IM, (i) TPFVE with our boundary conditions.

design, they are directly applicable to artwork development, textile and clothing design, and customized interior textures. Beyond aesthetics, the method also serves technical purposes: in game development, it supports seamless UV tiling for environmental textures such as grasslands and brick walls; in fashion design, it enables rapid generation of continuous fabric prints that scale naturally across garment surfaces, addressing a core demand in scalable textile design. Figure 22 demonstrates a representative set of patterns applied to garment design.

To evaluate practical efficiency, we measured computational performance. Thanks to GPU-parallelized core computations, the method achieves fast rendering without compromising textural detail or structural coherence. All experiments were performed using the parameter set $\{11.187, 15, 3.236, 4, 16, 16, 0.434, 0.434\}$. As summarized in Table III, pattern generation remains below one millisecond for all evaluated tiling schemes across resolutions ranging from 512×512 to 4096×4096 pixels. This real-time performance supports interactive design workflows, including rapid iteration, real-time preview, and immediate client communication, effectively bridging the gap between computational efficiency and creative demands.

TABLE III
RENDERING TIMES UNDER GPU ACCELERATION FOR PATTERNS OF DIFFERENT RESOLUTIONS AND TILING SCHEMES (MS)

	w/o tilings	k-uniform	Robinson	Chair	Quad
512×512	0.313	0.348	0.404	0.716	0.407
1024×1024	0.318	0.434	0.367	0.394	0.780
2048×2048	0.326	0.864	0.459	0.404	0.659
4096×4096	0.330	0.392	0.471	0.463	0.619
Average	0.322	0.510	0.425	0.494	0.616

VII. CONCLUSION

This paper has presented a physics-inspired paradigm for aesthetic pattern generation based on the Thin-Plate Free Vibration Equation (TPFVE). By establishing a principled mapping between the intrinsic characteristics of vibration modes and visual expression, our approach bridges the gap between mathematical rigor and artistic intuition. At the core of this paradigm lies a unified formulation in which both texture and layout emerge directly from a single analytical solution, with aesthetic constraints seamlessly encoded as boundary conditions. This eliminates the need for manually constructed IMs required by prior tiling-based methods, thereby fundamentally simplifying the creative workflow.

Experimental results confirm that the model provides rich morphological control through adjustable physical parameters and flexible boundary conditions, efficiently generating diverse patterns with structural integrity and detailed textures. Compared to existing methods, it demonstrates superior controllability, expressiveness, and computational efficiency, while offering a resource-efficient and interpretable alternative to data-driven generative models.

Although the method accommodates a wide range of periodic and aperiodic tilings, deriving analytical solutions for highly complex or curved boundaries remains a challenge, a limitation shared with traditional invariant mapping approaches. Future work will focus on extending the model to more general boundary formulations and exploring hybrid models that combine vibration-based modeling with complementary generative techniques, further advancing this interpretable and efficient pathway for digital art creation.

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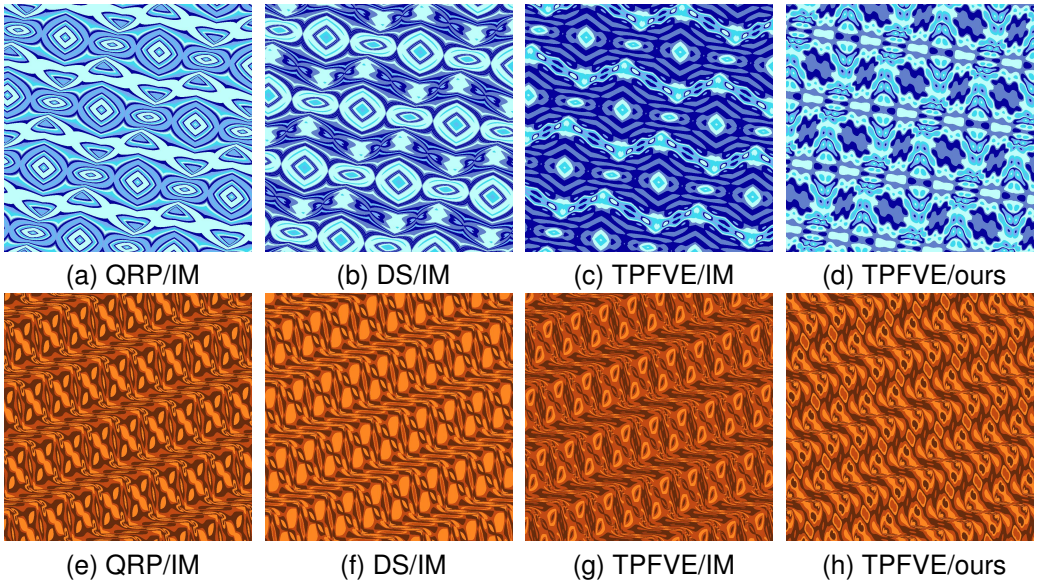

Fig. 21. Comparison with Wang et al. [14] on convex (a–d) and concave (e–h) quadrilateral tilings. Schemes: QRP+IM, DS+IM, TPFVE+IM, and TPFVE with our boundary conditions.

Fig. 22. Application of the aesthetic patterns generated by our method to garment design.

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